

Lab4 Report, Part A (Hard cases for Speech Recognition)

I tried the ASR with different words, accents, and languages.

Accent plays an important role in recognizing the right words with high confidence. I tried to mimic different English accents, and they brought the detection confidence down.

I tried to mimic the exact pronunciation of the TTS, and it didn't help. So, I conclude that the TTS and ASR are probably not trained jointly for each model. There were also cases where the ASR recognition was correct with a high percentage, but the TTS was not pronouncing the word properly, which confirms my conclusion. This mainly happened when I switched the language of the model to my mother tongue.

There are two major factors that can improve the accuracy and confidence of ASR:

- 1- The context: neighboring words and the syntax of whole expressions can drastically help ASR recognize individual words.
- 2- Precise and well-articulated production of consonants and vowels (in contrast with quick and more natural pronunciation) can help improve ASR recognition accuracy and confidence.

Another important factor in recognizing the correct word(s) is the possible existence of a minimal pair. In such cases, ASR may falter between the two, even with a high confidence score.